

Book A Doctor: Online Doctor Appointment Booking Platform

Project Description:

Book A Doctor is a web and mobile platform that enables patients to book doctor appointments online, view real-time availability, and receive automated reminders. The platform includes a Doctor Search & Booking module that lets patients find specialists by location and specialty, a Schedule Management module that keeps doctor availability updated in real time, and a Notification module that sends automated SMS and email reminders ahead of appointments.

The solution integrates secure online payments and provides a dedicated admin dashboard for hospitals to manage doctors, appointments, and users. By digitizing the booking process, Book A Doctor reduces patient waiting times, prevents double bookings, and improves overall healthcare accessibility for patients, doctors, and hospital administrators alike.

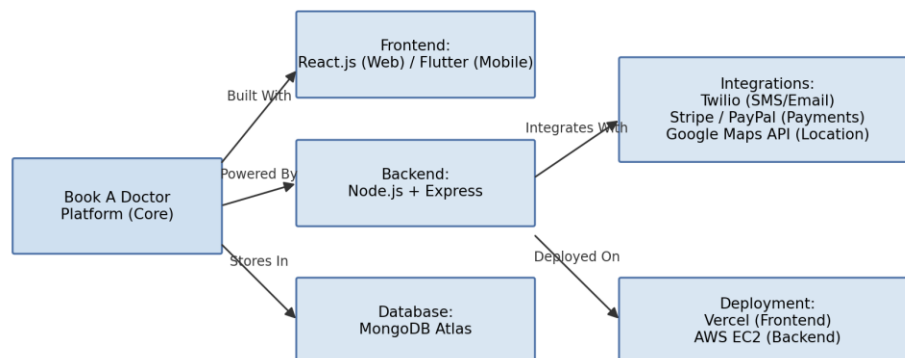
Scenarios

Scenario 1: A patient searches for a cardiologist in their city, views available time slots in real time, and books an appointment instantly, receiving an SMS and email confirmation.

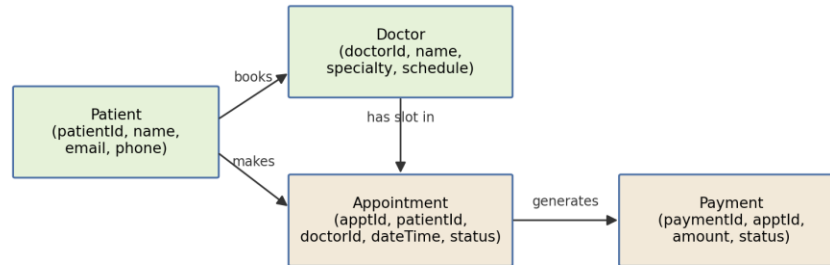
Scenario 2: A doctor updates their weekly schedule in the system, ensuring patients only see accurate, currently available slots and preventing double bookings.

Scenario 3: A hospital admin logs into the dashboard and generates a weekly report of appointments and payments, giving the hospital visibility into booking trends and revenue.

Technical Architecture:



Description: Book A Doctor uses a modular client-server architecture. The frontend is built with React.js for the web application and Flutter for the mobile app, giving patients and doctors a responsive, cross-platform experience. The Node.js and Express backend handles authentication, booking logic, and orchestrates calls to third-party integrations, while all application data — patients, doctors, appointments, and payments — is persisted in MongoDB Atlas. The platform integrates the Twilio API for SMS/email reminders, Stripe/PayPal for secure payments, and the Google Maps API for clinic location lookup. The frontend is deployed on Vercel and the backend on AWS EC2, with Ngrok used for public access during local testing.



Database Description (ER Diagram): The data model centers on four core collections. A Patient books an Appointment with a Doctor, who maintains their own schedule; each Appointment can generate a Payment record once the booking is confirmed. This structure keeps booking, scheduling, and billing data cleanly related while remaining simple to query and scale in MongoDB.

Pre-requisites:

- **Node.js Runtime (v18+):** <https://nodejs.org/en/docs>
- **React.js Knowledge:** <https://react.dev/learn>
- **Flutter Knowledge:** <https://docs.flutter.dev/>
- **MongoDB Atlas Familiarity:** <https://www.mongodb.com/docs/atlas/>
- **Twilio API Familiarity:** <https://www.twilio.com/docs>
- **Payment Gateway (Stripe) Documentation:** <https://stripe.com/docs>
- **Version Control with Git:** <https://git-scm.com/doc>
- **Vercel Deployment Guide:** <https://vercel.com/docs>
- **AWS EC2 Documentation:** <https://docs.aws.amazon.com/ec2/>

Project Workflow:

Milestone 1: Requirement Gathering & Architecture Design

- **Activity 1.1:** Gather functional requirements for patients, doctors, and hospital admins.
- **Activity 1.2:** Select the technology stack: React.js/Flutter for frontend, Node.js/Express for backend, MongoDB Atlas for the database.
- **Activity 1.3:** Define the system architecture, detailing interactions between the frontend, backend, database, and third-party integrations.
- **Activity 1.4:** Set up the development environment, installing Node.js, MongoDB tools, and required dependencies.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop patient registration and login functionality with secure authentication.
- **Activity 2.2:** Build the doctor search and appointment booking flow, including availability checks.
- **Activity 2.3:** Implement real-time schedule management so doctors can update availability and prevent double bookings.

Milestone 3: Backend Services

- **Activity 3.1:** Implement authentication and authorization for patients, doctors, and admins.
- **Activity 3.2:** Integrate the Twilio API to send automated SMS and email appointment reminders.
- **Activity 3.3:** Integrate Stripe/PayPal for secure payment processing on appointment booking.

Milestone 4: Frontend Development

- **Activity 4.1:** Design and build a responsive UI/UX for the web (React.js) and mobile (Flutter) apps.
- **Activity 4.2:** Build the admin dashboard for hospitals to manage doctors, appointments, and users.
- **Activity 4.3:** Integrate the Google Maps API so patients can view clinic locations.

Milestone 5: Deployment & Testing

- **Activity 5.1:** Test the application locally, verifying booking, notification, and payment flows end-to-end.
- **Activity 5.2:** Deploy the frontend to Vercel and the backend to AWS EC2, using Ngrok for interim public testing.
- **Activity 5.3:** Run performance and load testing to validate the system under expected and peak traffic.

Milestone 6: Enhancements

- **Activity 6.1:** Add multi-language support to widen accessibility for non-English-speaking patients.
- **Activity 6.2:** Build a telemedicine video consultation module for remote doctor visits.

Conclusion:

Book A Doctor is a full-stack healthcare platform built with React.js, Flutter, Node.js, and MongoDB Atlas that streamlines the entire appointment booking journey. It combines real-time schedule management, automated Twilio-powered reminders, and secure payment processing with a hospital-facing admin dashboard, reducing waiting times and preventing double bookings. The project spans requirement gathering, core feature development, backend services, frontend design, and cloud deployment, with performance testing confirming stable behavior under load. Looking ahead, planned enhancements such as multi-language support and telemedicine video consultations position Book A Doctor to grow from a booking tool into a comprehensive digital front door for healthcare providers and their patients.